Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2026

Course: B.Tech
Code: 101405

Subject: Structural Analysis

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) The degree of static indeterminacy of a propped cantilever beam subjected to vertical loads is
(i) 0
(ii) 1
(iii) 2
(iv) 3

(b) The unit load method is primarily used to determine
(i) The degree of static indeterminacy
(ii) Support reactions only
(iii) Displacements and rotations in structures
(iv) The yield strength of structural materials

(c) The principle of virtual work states that a system is in equilibrium when the total virtual work done by external and internal forces is
(i) Maximum
(ii) Minimum
(iii) Zero
(iv) Equal to the total strain energy in every case

(d) The Müller-Breslau principle can be applied to determine qualitative influence lines for
(i) Only statically determinate beams
(ii) Only statically indeterminate frames
(iii) Both statically determinate and indeterminate structures
(iv) Only simply supported beams

(e) To obtain the qualitative influence line for the bending moment at an intermediate section of a beam, one should
(i) Release the vertical support at that section
(ii) Release the rotational restraint at that section and impose a unit rotation
(iii) Apply a unit axial displacement at the section
(iv) Remove the entire beam section without imposing any displacement

(f) The moment-area method is based primarily on the relationship between the
(i) Shear force and bending moment diagrams
(ii) Bending moment and curvature of the elastic curve
(iii) Axial force and shear force
(iv) Load intensity and cross-sectional area

(g) For a prismatic member with the far end fixed, the carry-over factor is
(i) 0
(ii) 1/4
(iii) 1/2
(iv) 1

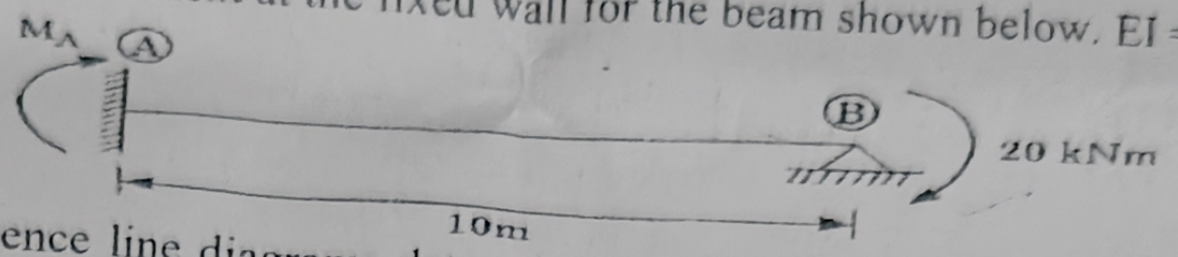
(h) In the moment distribution method, sway correction is required when
(i) The frame undergoes lateral joint translation
(ii) All joints are prevented from translating laterally
(iii) The beam carries only a uniformly distributed load
(iv) The structure has no rotational degrees of freedom

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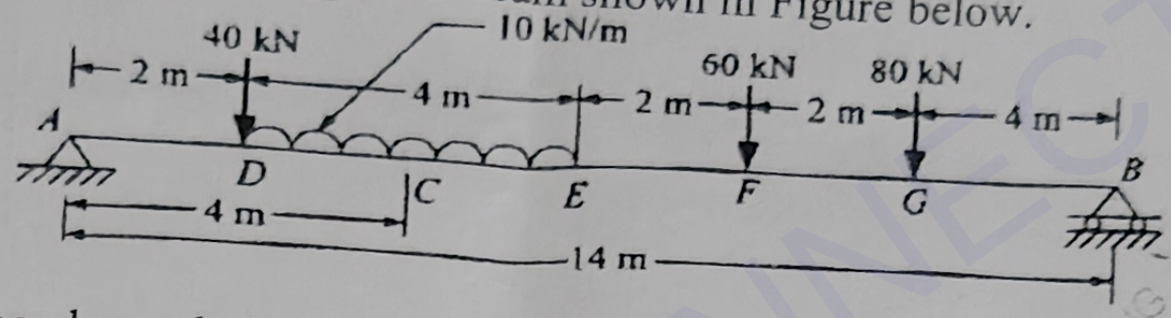
(i) The fundamental matrix equation used in the direct stiffness method of structural analysis is
(i) $[F]\{P\} = \{\Delta\}$
(ii) $[K]\{\Delta\} = \{P\}$
(iii) $[P]\{K\} = \{\Delta\}$
(iv) $[K]\{P\} = \{\Delta\}$

(j) In the displacement method of structural analysis, the primary unknowns are
(i) Member forces only
(ii) Joint displacements and rotations
(iii) Support reactions only
(iv) Material densities and unit weights

Q.2 (a) Determine the moment at the fixed wall for the beam shown below. EI = constant. [7]



(b) Using influence line diagrams determine the shear force and bending moment at section C in the simply supported beam shown in Figure below. [7]



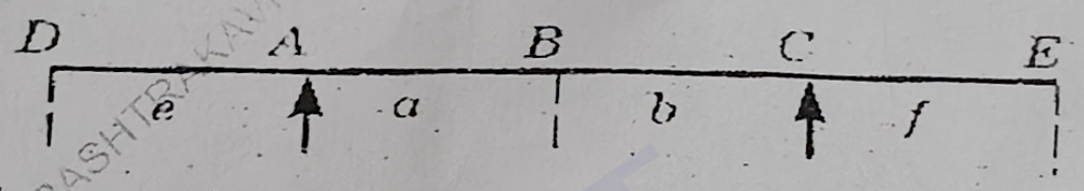
Q.3 (a) A three-hinged parabolic arch of span 20 metre and rise 4 metre carries a uniformly distributed load of 20 kN per metre run on the left half of the span. Find the maximum bending moment for the arch. [7]

(b) A beam of span 10 m is fixed at each end and carries a downward point load of 70 kN at a distance of 3 m from the left end and an upward point load of 30 kN at midspan. Find (i) the reactions at each support and (ii) The bending moment at each support. Draw also B.M. and S.F. diagrams. [7]

Q.4 (a) A beam of span l carries a concentrated load P at mid-span. Find the central deflection using Strain Energy Method. [7]

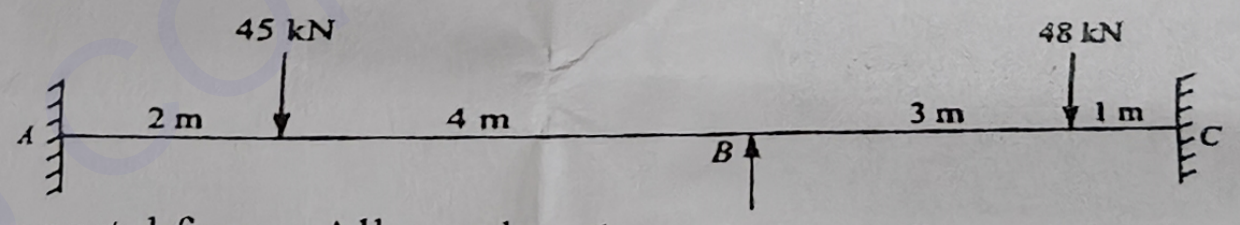
(b) Find the central deflection of a simply supported beam carrying a concentrated load at mid-span. Assume uniform flexural rigidity. Use Castigliano's theorem. [7]

Q.5 (a) Draw ILD for R_A, V_B & M_B. [7]

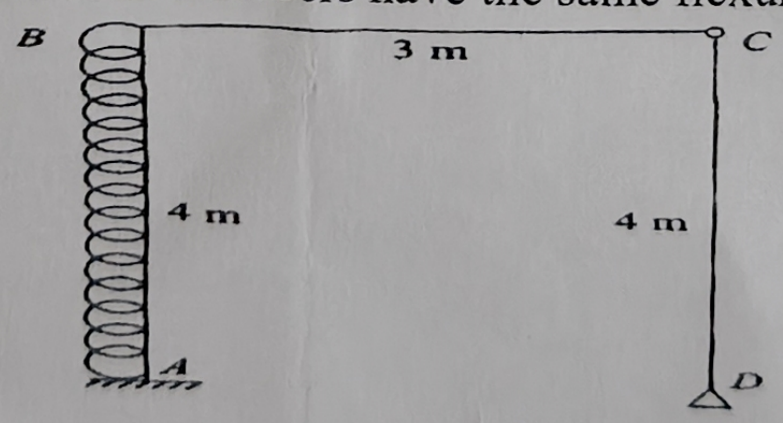


(b) A continuous beam consists of three successive spans of 8 metres, 10 metres and 6 metres and carries loads of 60 kN per metre, 40 kN per metre and 80 kN per metre respectively on the spans. Determine the bending moments and reaction at the supports. [7]

Q.6 (a) Analyse the continuous beam using slope deflection method. Draw also B.M. and S.F. diagrams. [7]

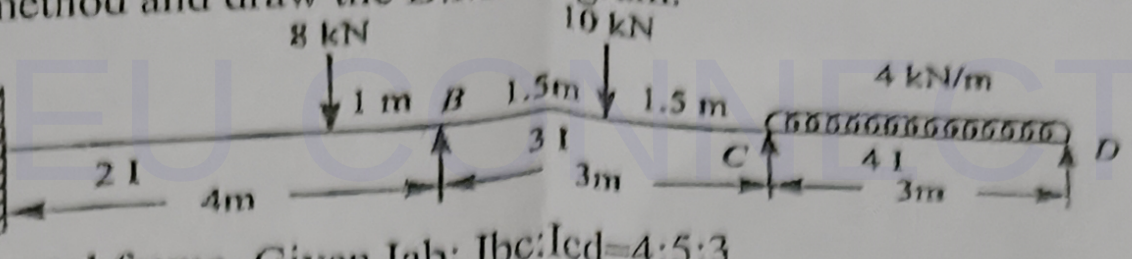


(b) Analyse the portal frame. All members have the same flexural rigidity. [7]

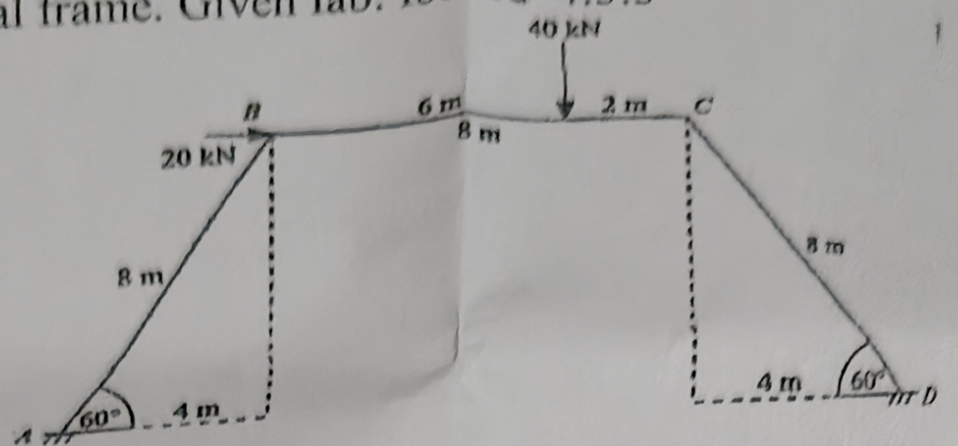


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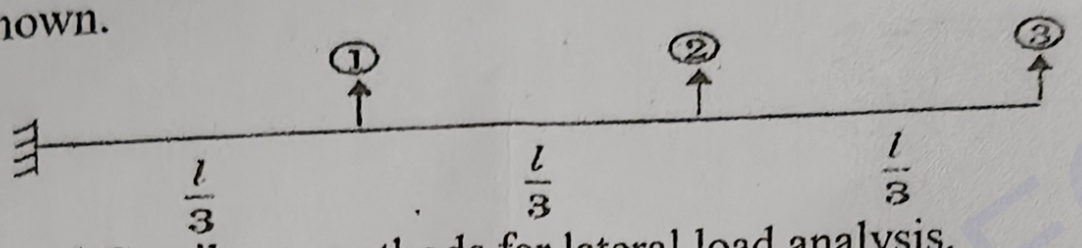
Q.7 (a) Determine the support moments for the continuous beam using moment distribution method and draw the B.M. diagram. [7]



(b) Analyse the portal frame. Given Iab: Ibc:Icd=4:5:3 [7]



Q.8 (a) Develop flexibility matrix for the beam shown in figure below with respect to the Co-ordinate shown. [7]



(b) Explain portal and Cantilever methods for lateral load analysis. [7]

[7 x 2 = 14]

Q.9 Write short note on ***any two*** of the following:

(a) Maximum bending moment under a chosen wheel load
(b) Analysis of Beams and Frames: Moment Area method
(c) Effect of symmetry and ant symmetry
(d) Lateral load analysis:Displacement/Stiffness methods

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